AutoTeaKG-Silver: A Validated Silver-Standard Evidence Graph for Tea Functional Activity Literature

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Abstract

Tea functional-activity literature is difficult to synthesize because tea type, component identity, processing state, extraction method, study design, endpoint label, and mechanism evidence are often reported in incompatible forms. AutoTeaKG-Silver addresses this gap by converting PubMed tea bioactivity records into a provenance-rich, uncertainty-aware evidence graph. The pipeline retrieves PubMed records, extracts structured title/abstract and open full-text methods evidence, normalizes activity and evidence labels, recovers processing and extraction context, and exports graph edges through explicit evidence-record nodes. In the current KG v3 release, AutoTeaKG-Silver contains 635 evidence records, 1,989 nodes, and 8,195 edges. Targeted extraction increased processing-context coverage from 146 to 183 records and extraction-context coverage from 154 to 185 records, while retaining missing-context flags for 303 records. External validation of 47 completed stratified records showed acceptable rates of 92% for activity category, 80% for study type, 86% for evidence level, 94% for component group, 100% for processing step, 100% for extraction method, and 94% for mechanism label. The resource exposes uneven evidence support across activity classes and enables constrained microbiome-mechanism queries that preserve evidence level and uncertainty. AutoTeaKG-Silver is therefore intended as a reproducible silver-standard infrastructure for living tea bioactivity synthesis, not as a manually verified gold-standard database.

Database URL: https://github.com/ShawnXiha/AutoTeaKG_Silver

1 Introduction

Tea is a chemically complex beverage and botanical material with a large literature on potential health-related functions. Reviews have summarized the biological potential of catechins, theaflavins, theanine, caffeine, polysaccharides, and other tea-derived compounds across antioxidant, inflammatory, metabolic, cardiovascular, and neuroprotective endpoints [1–3]. A parallel literature studies how tea processing and extraction alter chemical profiles and product quality [4–6]. More recently, tea–microbiome interactions have become a prominent mechanistic theme, especially for metabolic and gut-brain-axis phenotypes [7–9]. These bodies of work make tea an attractive domain for structured evidence synthesis.

The central difficulty is that tea functional activity evidence is abundant but poorly aligned. A claim about "green tea", "EGCG", "tea polyphenols", "fermented tea", or "tea polysaccharides" may refer to different materials, processing states, extraction methods, doses, and experimental settings. Evidence strength also varies substantially: animal studies, in vitro experiments, randomized trials, cohort studies, and narrative reviews often appear together in the same prose summary. Narrative reviews are useful for expert interpretation, but they do not provide a machine-readable evidence layer that can answer comparative questions such as which activities have human evidence, which are mostly preclinical, or whether processing information is reported at all.

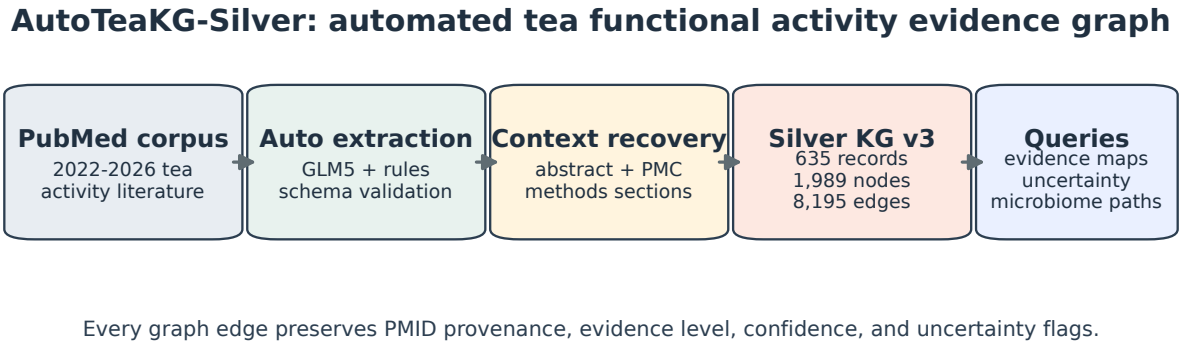

Figure 1: Graphical overview of AutoTeaKG-Silver. The pipeline converts PubMed tea functional activity literature into structured evidence records, recovers processing and extraction context from abstracts and open PMC methods sections, and exports a provenance-rich uncertainty-aware evidence graph.

Existing tea databases do not solve this functional evidence problem. TeaPVs and TeaNekT provide genomic and transcriptomic resources for tea plant biology [10, 11]. TRSRD is closer to a tea-domain knowledge graph, but it focuses on risky substances rather than functional activity evidence [12]. More general natural-product resources such as NPASS and natural-product drug-interaction knowledge graphs show that natural products can be represented computationally [13, 14], but they do not encode the tea-specific combination of functional activity, processing context, evidence level, and microbiome mechanism.

We argue that the missing contribution is not another static spreadsheet of tea papers, but an automated evidence infrastructure that makes uncertainty explicit. Manual curation can improve record quality, but it does not scale to a living PubMed-updated resource. Conversely, raw large-language-model extraction is not sufficient unless outputs are constrained by domain schemas, provenance, validation flags, and uncertainty classes. The practical goal is therefore to build a silver-standard graph: a resource that is automatically generated, reproducible, inspectable, and useful for evidence mapping, while avoiding the stronger claim that every edge is expert verified.

This paper introduces AutoTeaKG-Silver, an uncertainty-aware evidence graph for tea functional activity literature. The pipeline retrieves PubMed records, generates structured evidence records, extracts tea type, component group, activity category, study type, evidence level, mechanism labels, microbiome features, and host phenotypes, and then enriches records with processing and extraction context from abstracts and open full-text methods sections. The final graph preserves evidence-record nodes so that each relation can be traced back to a paper, raw claim text, evidence level, and uncertainty flag.

The contributions are fourfold and each is tied to a result-level evidence anchor (Table 1). First, we define a graph-ready evidence model that separates activity claims from study design, evidence level, component context, processing context, and uncertainty. Second, we build an automated silver-standard resource with 635 evidence records and 8,195 graph edges, excluding manual annotation from the data-generation path. Third, we show that processing and extraction context remain a measurable reporting bottleneck: targeted extraction improves coverage, but many records lack open full text or sufficient methods detail. Fourth, we provide query tables and graph outputs that support evidence mapping, processing-aware analysis, and microbiome mechanism exploration

Table 1: Claim–evidence map for the main manuscript. Each central claim is tied to at least one table or figure anchor.

Claim	Supporting evidence	Anchor
The resource is a graph rather than a flat bibliography.	635 evidence records, 1,989 nodes, and 8,195 edges with explicit EvidenceRecord provenance.	Table 4; Figure 2
Evidence support differs across activity classes.	Activity-by-evidence heatmap separates preclinical, synthesis, observational, and interventional support.	Figure 4; Figure 2
Processing and extraction context remain a reporting bottleneck.	Context recovery improves across extraction stages, but 303 records retain missing-context flags.	Table 5; Figure 2
The graph supports constrained mechanism queries.	A tea-polysaccharide query returns microbiome, metabolite, phenotype, evidence, and uncertainty-linked paths.	Table 6; Figure 9
Human validation supports silver-standard use while identifying weak fields.	Acceptable rates range from 80% for study type to 100% for processing and extraction fields.	Table 7; Figure 2

under explicit uncertainty filters.

1.1 Prior work and resource gap

1.1.1 Tea bioactivity reviews and evidence synthesis

The tea bioactivity literature is rich in narrative synthesis. Recent reviews summarize the biological potential of tea bioactive compounds across inflammation, oxidation, metabolism, microbiota modulation, and other health-related endpoints [1]. Other reviews focus on specific mechanistic constraints, such as the bioavailability of tea polyphenols and the difficulty of extrapolating in vitro activity to in vivo health effects [2]. Tea polysaccharides have also become a distinct review topic because their chemical characterization, conjugates, and bioactivities differ from those of catechin-centered polyphenol literature [3]. These reviews establish that the domain is active and biologically broad, but their outputs remain prose summaries rather than computable evidence records. AutoTeaKG-Silver differs by representing study type, evidence level, activity category, component context, and uncertainty as queryable fields.

1.1.2 Processing-aware tea chemistry and microbiome mechanisms

Tea processing and extraction strongly influence chemical composition, product quality, and downstream interpretation. Food engineering reviews describe ultrasound, microwave, pressurized-liquid, supercritical, and other extraction technologies for recovering tea bioactives [4]. Tea processing reviews further show that harvesting, withering, rolling, fermentation or oxidation, drying, and storage can alter phytochemical profiles, especially in black tea and fermented products [5]. Metabolomics studies provide analytical evidence that processing technology changes tea quality and chemical profiles [6]. In parallel, microbiome-focused work links differently processed tea, tea polyphenols, fermented tea, and tea polysaccharides to gut microbiota, microbial metabolites, host barrier function, metabolic outcomes, and cognition-related phenotypes [7–9, 15, 16]. These studies motivate the processing-aware and microbiome-aware dimensions of AutoTeaKG-Silver.

1.1.3 Tea databases, natural-product resources, and knowledge graphs

Existing tea resources address important but different layers of tea biology. TeaPVs focuses on genomic variation in *Camellia sinensis*, and TeaNekT provides comparative transcriptomic information for tea plant biology [10, 11]. TRSRD uses natural-language processing and knowledge graph techniques for tea risky-substance research, but the target domain is risk and safety rather than functional activity evidence [12]. Broader natural-product resources such as NPASS provide quantitative activity and species-source information, and natural-product drug-interaction knowledge graphs show the feasibility of graph-based representation for natural products [13, 14]. AutoTeaKG-Silver is complementary to these resources: it targets tea functional activity claims, evidence strength, processing/extraction context, microbiome mechanisms, and uncertainty-aware provenance.

2 Materials and Methods

2.1 Corpus construction and retrieval archive

The corpus was constructed from PubMed searches over tea functional activity, component, processing, extraction, microbiome, and human-evidence query blocks. The archived query set covered the publication window from 2022-01-01 to 2026-03-31 and included broad review retrieval, functional activity primary studies, microbiome-focused studies, processing-aware studies, and human evidence searches. Across the larger retrieval batches, records were deduplicated by PMID and normalized into paper-level metadata tables. The manuscript uses the locked auto-only evidence graph derived from these retrievals rather than manually adjudicated mini-gold records.

2.2 Evidence-record schema

Each evidence record represents a structured claim anchored to a source paper. Core fields include the source paper identifier, tea type, material form, component group, activity category, endpoint label, study type, evidence level, model system, dose or exposure text, effect direction, mechanism label, microbiota taxon, microbial metabolite, host phenotype, raw claim text, confidence score, and provenance metadata. Activity categories include antioxidant, anti-inflammatory, metabolic improvement, anti-obesity, gut microbiota modulation, neuroprotection, cardiovascular protection, and other. Evidence levels distinguish preclinical in vivo evidence, human observational evidence, human interventional evidence, evidence synthesis, non-quantitative evidence synthesis, and low-preclinical evidence.

2.3 Auto-only extraction pipeline

AutoTeaKG-Silver was designed as an auto-only pipeline. The construction path used PubMed metadata, title/abstract text, candidate-record heuristics, GLM5 title/abstract extraction, schema validation, post-processing, duplicate consolidation, and graph export. Manual audit worksheets, mini-gold annotations, and adjudication logs were excluded from the data-generation path. This design choice is deliberate: the goal is to create a scalable silver-standard evidence layer whose uncertainty is visible, rather than to claim expert-level verification for each relation.

Table 2: Compact construction-stage summary. Detailed scripts and output paths are listed in Appendix A.

Stage	Output	Count or result
PubMed retrieval and normalization	archived query batches	5 query blocks
Auto-only evidence extraction	silver evidence records	635 records
Abstract-level context recovery	processing/extraction patches	365 patches
PMC methods retrieval	methods-like sections	151 records
Methods-level context recovery	methods-section patches	151 patches
Vocabulary-normalized KG v3	graph and figure tables	1,989 nodes; 8,195 edges

2.4 Processing and extraction context recovery

Processing and extraction context were recovered in three stages. The first stage used conservative rule-based matching over titles, abstracts, and claim text. The second stage targeted records flagged as missing processing or extraction context and prompted GLM5 to extract only tea-material processing, extraction, and component context from title/abstract text. The third stage queried NCBI E-utilities to map PMIDs to open PMC full text, extracted methods-like sections, and ran a targeted methods-section extractor on records with available full-text context. All targeted extraction steps were patch-first: the model generated a patch table, and patched copies of the silver dataset and KG were regenerated rather than overwriting the base resource.

2.5 Vocabulary normalization

LLM surface labels were normalized into a controlled processing and extraction vocabulary. For example, black tea manufacturing and black tea formation were mapped to black tea processing; brick-like form and brick tea processing were mapped to compressed/brick tea processing; cold brewing and hot brewing were mapped to brewing/infusion; and isolation, isolated, and purification were mapped to isolation/purification. Raw LLM labels were retained in mapping tables so that normalization decisions remain auditable.

2.6 Uncertainty assignment

AutoTeaKG-Silver assigns uncertainty at the evidence-record level and propagates the same attributes to graph edges through the evidence-record node. The silver confidence score combines the original extraction confidence with record specificity and context completeness. Context completeness assigns partial credit for non-generic tea type, component group, processing step, and extraction method, while mechanism specificity distinguishes missing or generic mechanisms from pathway-like or named mechanism labels. Records are then assigned uncertainty flags (Appendix Table 9) and mapped to low, moderate, or high uncertainty classes using the rules in Table 3. This design keeps uncertainty queryable rather than hiding it during filtering.

2.7 Graph construction and query tables

The graph preserves papers and evidence records as explicit nodes. Evidence records are linked to tea type, material form, component group, processing step, extraction method, activity category, study type, evidence level, effect direction, mechanisms, microbiota features, metabolites, host phenotypes, and uncertainty flags. Graph edges retain paper identifier, record identifier, evidence level, study type, effect direction, confidence score, and uncertainty class. Query tables were

Table 3: Uncertainty-class assignment rules. The severe flag set includes low LLM confidence, taxonomy expansion needed, missing component context, generic mechanism, and invalid controlled-vocabulary labels.

Class	Rule	Final count
Low uncertainty	Silver confidence ≥ 0.80 and no severe uncertainty flag.	100
Moderate uncertainty	Silver confidence ≥ 0.60 and at most two severe uncertainty flags.	470
High uncertainty	All remaining records.	65

generated from the final KG v3 to support evidence mapping, activity-by-evidence heatmaps, uncertainty summaries, processing and extraction label distributions, microbiome mechanism subsets, and node/edge composition summaries.

2.8 External validation

To avoid using manual review as a hidden construction dependency, validation was separated from graph generation. We prepared a stratified validation sample of 48 auto-only evidence records: 14 low-uncertainty records, 24 moderate-uncertainty records, and 10 high-uncertainty records. Reviewers assessed activity category, study type, evidence level, component group, processing step, extraction method, and mechanism label as correct, minor issue, major issue, or not applicable. One sample remained incomplete, producing a final reviewed set of 47 records. The current KG v3 was not retrained or modified using these validation judgments.

2.9 Reproducibility package

The current manuscript release is organized so that each result can be traced to a script and output directory (Table 2). The final graph used for all Results figures is the vocabulary-normalized methods-level KG v3. The corresponding record table, node table, edge table, query tables, and figure-generation script are listed in Appendix A.

3 Results

3.1 AutoTeaKG-Silver constructs a provenance-rich tea functional activity graph

The final KG v3 contains 635 silver evidence records, 1,989 nodes, and 8,195 edges (Table 4; Figure 2). The graph includes paper, evidence-record, activity-category, study-type, evidence-level, tea/context, mechanism, microbiota, metabolite, phenotype, and uncertainty nodes. The explicit EvidenceRecord layer is the central design choice: it prevents mechanistic and activity relations from becoming detached from source papers, raw claims, evidence levels, and uncertainty classes.

3.2 Evidence-level normalization exposes uneven support across activity classes

The activity-by-evidence heatmap shows that tea functional activity claims are not evenly supported across evidence levels (Figure 4). Several activity categories are dominated by preclinical in vivo records and non-quantitative evidence syntheses, while human observational and interventional evidence is concentrated in fewer categories. This pattern supports the central motivation

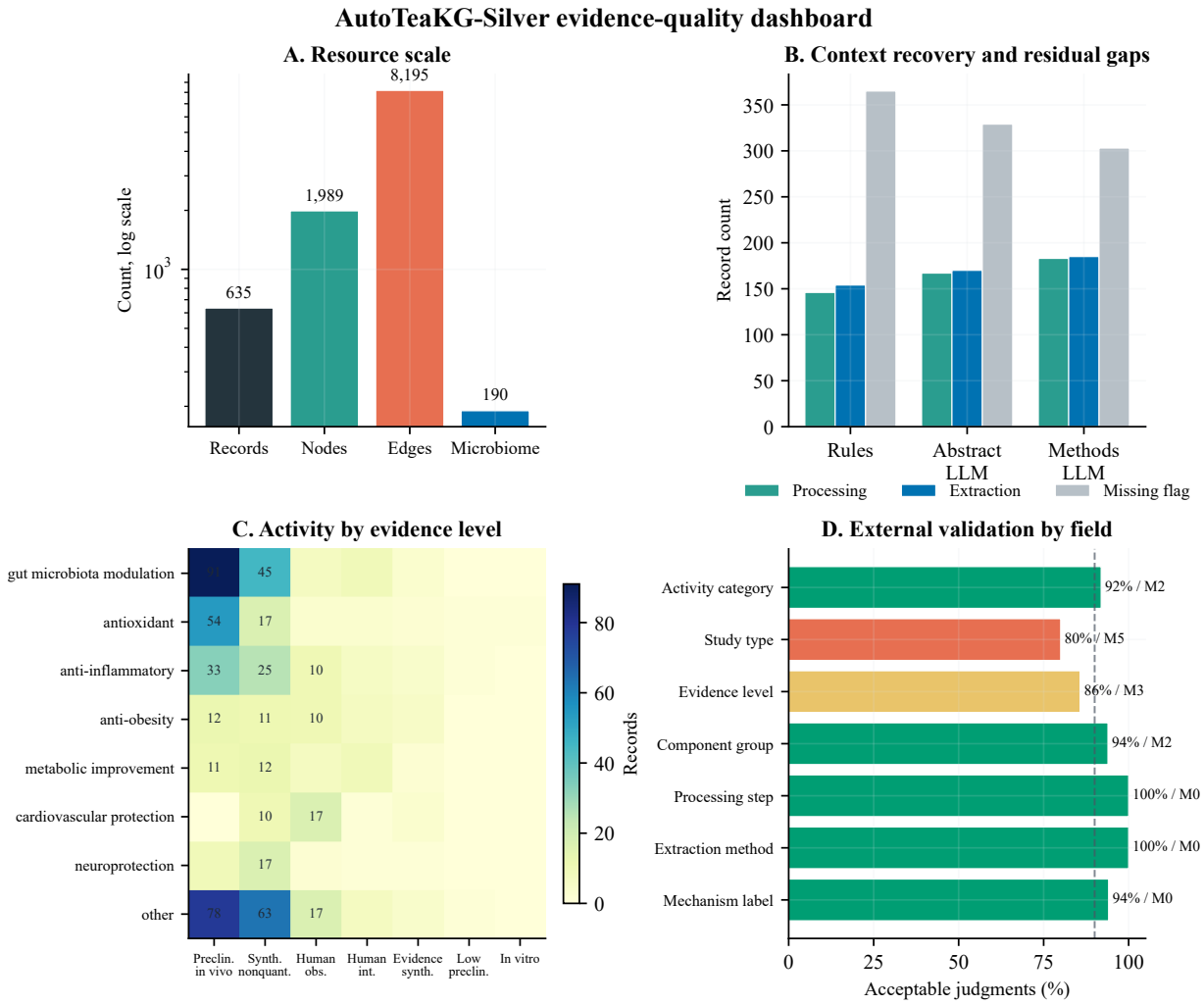

Figure 2: Evidence-quality dashboard for AutoTeaKG-Silver. The figure summarizes resource scale, context recovery, activity-by-evidence structure, and external validation rates in one reproducible anchor figure.

for evidence grading: without evidence-level normalization, review-derived, animal, cohort, and randomized-trial claims are easily mixed into a single apparent evidence base.

3.3 Targeted extraction improves but does not eliminate missing processing context

Processing and extraction context increased across extraction stages (Figure 5). Rule-based extraction identified processing context in 146 records and extraction context in 154 records. Abstract-level targeted GLM5 extraction increased coverage, and PMC methods-section extraction further raised processing-context coverage to 183 records and extraction-context coverage to 185 records. The improvement is measurable but modest, indicating that processing context is not uniformly available from abstracts alone.

The stage-wise quality-control summary shows that each pipeline stage contributes measurable context recovery while preserving graph integrity (Table 5). Processing-context coverage increased

Table 4: Final KG v3 summary statistics used throughout the Results section.

Quantity	Count
Silver evidence records	635
KG nodes	1,989
KG edges	8,195
Microbiome-relevant records	190
Records with processing context	183
Records with extraction context	185
Low-uncertainty records	100
Moderate-uncertainty records	470
High-uncertainty records	65
Residual missing-context records	303

Table 5: Stage-wise ablation and quality-control summary. Each stage adds context or normalizes previous outputs while preserving a zero unresolved-edge graph export. Proc., processing-context records; Extr., extraction-context records; Missing, records still flagged for missing processing or extraction context.

Stage	Proc.	Extr.	Missing	Unmapped	Errors	Bad edges
Rules-only silver layer	146	154	365	–	–	0
Abstract-level targeted LLM	167	170	329	0	0	0
Methods-section targeted LLM	183	185	303	0	0	0

from 146 records in the rules-only silver layer to 167 records after abstract-level targeted extraction and 183 records after methods-section extraction. Extraction-context coverage increased from 154 to 170 and then to 185 records. The missing-context flag decreased from 365 to 303 records, while the final normalized KG retained zero unmapped processing/extraction labels and zero unresolved graph edges.

The full-text retrieval analysis clarified why context remains missing. Of 329 records still flagged after abstract-level extraction, 151 records had PMC methods-like sections, while 178 lacked open PMC full text (Figure 6). Methods-section extraction generated patches for all 151 available records and recovered details such as crushing and sieving, cleaning–drying–powdering, steaming and fixation, ultrasonic ethanol extraction, methanol sonication, microwave-assisted extraction, brewing, and isolation/purification.

3.4 Vocabulary normalization reduces processing and extraction label fragmentation

Processing and extraction labels generated by LLM extraction required normalization before graph use. The final vocabulary-normalized KG reduced surface-form fragmentation while retaining raw labels for provenance. Fermentation/oxidation was the dominant processing context, followed by storage/aging, enzymatic treatment, roasting, black tea processing, processing category contrast, drying, steaming/fixation, and crushing/grinding/sieving. Extraction labels were concentrated in other extraction, aqueous extraction, isolation/purification, analytical characterization, brewing/infusion, solvent extraction, ultrasound-assisted extraction, and commercial preparation (Figure 7).

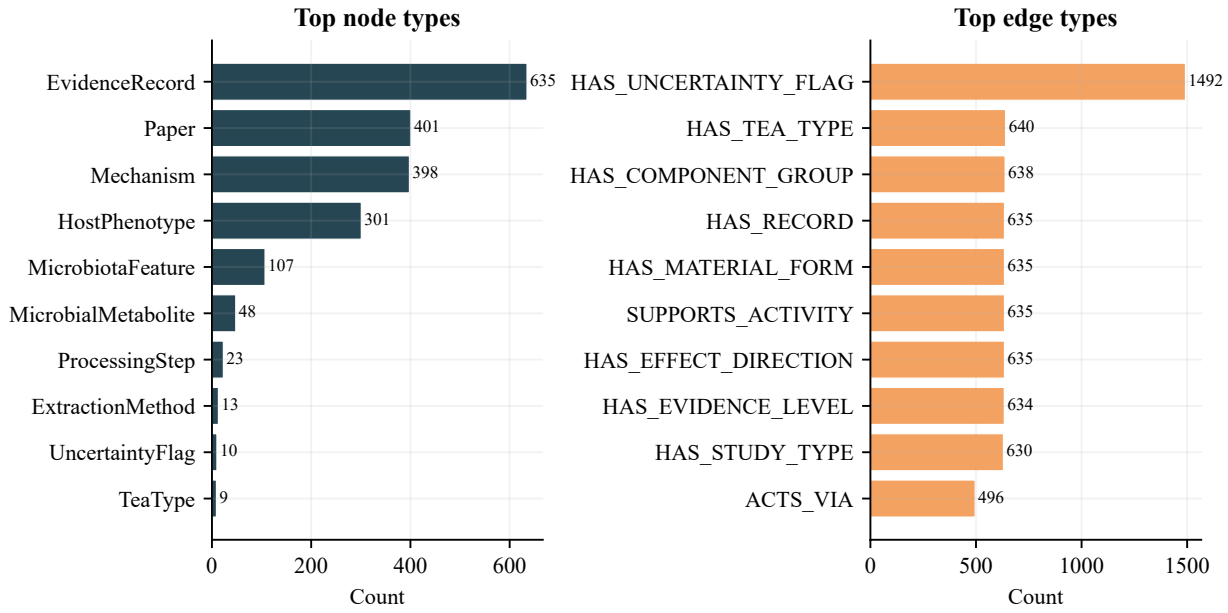

Figure 3: Node and edge type composition of KG v3. The graph preserves evidence-record provenance while linking papers, activity categories, evidence levels, tea/component context, processing/extraction context, mechanisms, microbiota features, metabolites, phenotypes, and uncertainty flags.

3.5 Uncertainty classes make silver-standard status explicit

The final KG contains 100 low-uncertainty records, 470 moderate-uncertainty records, and 65 high-uncertainty records. The activity-level uncertainty composition (Figure 8) shows that moderate uncertainty dominates the resource. This result is not a defect of the resource; it is the intended operating mode of a silver-standard evidence graph. Instead of filtering uncertainty away, AutoTeaKG-Silver stores uncertainty as a queryable property that downstream users can use to restrict analyses.

3.6 The graph supports microbiome-mechanism queries

KG v3 contains 190 microbiome-relevant records. These records connect tea types and component groups to gut microbiota modulation, microbial metabolites, host phenotypes, and mechanism labels. The microbiome subset provides a graph-native substrate for querying component–microbiota–metabolite–phenotype paths rather than treating microbiome mentions as isolated keywords. This design is motivated by recent work linking tea polyphenols, oolong tea polyphenols, and tea polysaccharides to microbiota-mediated metabolic, inflammatory, and neuroprotective programs [7–9, 15, 16].

3.7 A graph query recovers a tea-polysaccharide gut-liver axis pattern

To test whether the KG supports structured queries beyond flat keyword retrieval, we queried records with tea-polysaccharide component context and microbiome, metabolite, or host-phenotype fields. The query recovered five representative records linking tea polysaccharides to gut microbiota modulation, butyric acid or SCFA-related metabolites, and liver, barrier, or lipid-deposition pheno-

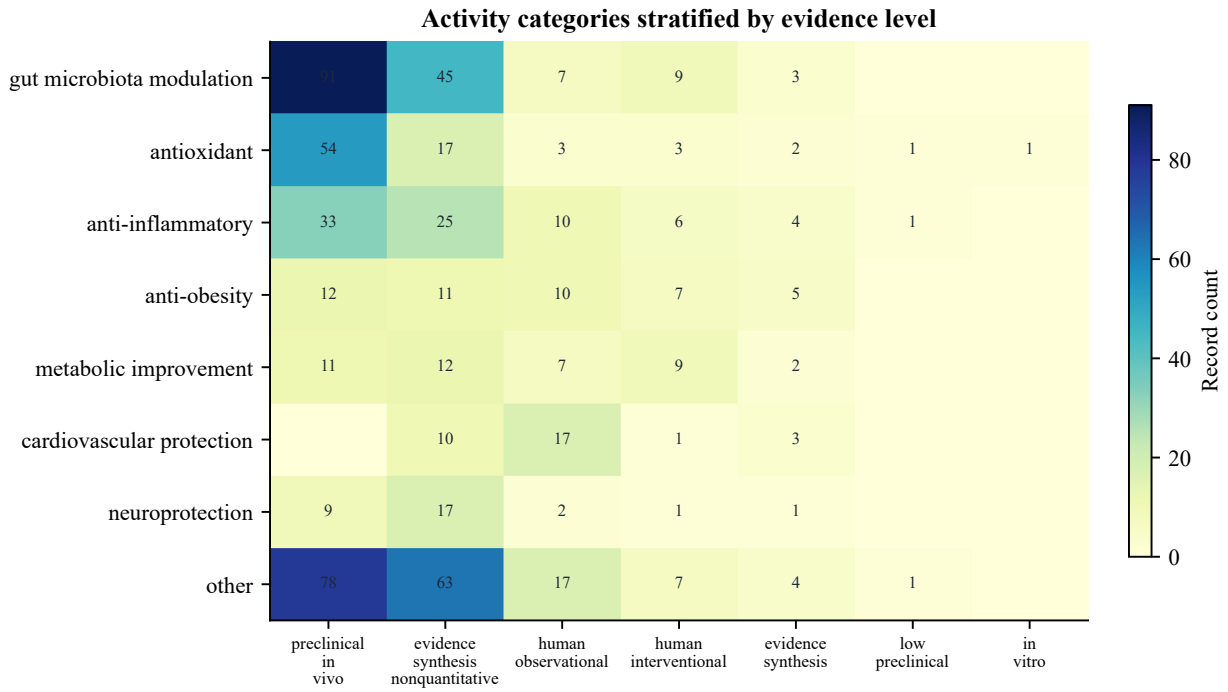

Figure 4: Heatmap of normalized activity categories stratified by evidence level in KG v3. The distribution shows that animal evidence and non-quantitative evidence syntheses dominate several activity categories, while human interventional and observational evidence is concentrated in fewer endpoints.

types (Table 6; Figures 9 and 10). The retrieved paths preserve evidence level and uncertainty class, allowing users to separate low-uncertainty preclinical gut-liver-axis records from broader moderate-uncertainty microbiome records. This case is the strongest utility test in the manuscript because the answer requires simultaneous constraints on component, mechanism, metabolite, phenotype, evidence, and uncertainty fields.

3.8 External validation supports the reliability of context-heavy fields

The completed 47-record validation sample provides an external check on the silver-standard graph (Table 7; Figure 2). The strongest fields were those most directly tied to context recovery: processing step and extraction method were correct in all completed judgments, while component group and mechanism label achieved acceptable rates of 94% or above. Activity category also remained strong with a 92% acceptable rate. Study type and evidence level were the main residual error sources, with acceptable rates of 80% and 86%, respectively. Major issue tags were concentrated in out-of-scope records, wrong activity labels, and wrong study-type assignments. These results support silver-standard use while identifying the fields that should be prioritized in future correction cycles.

4 Discussion

AutoTeaKG-Silver reframes tea functional activity synthesis as an automated evidence-graph construction problem. The main contribution is not that the system replaces expert review. Rather,

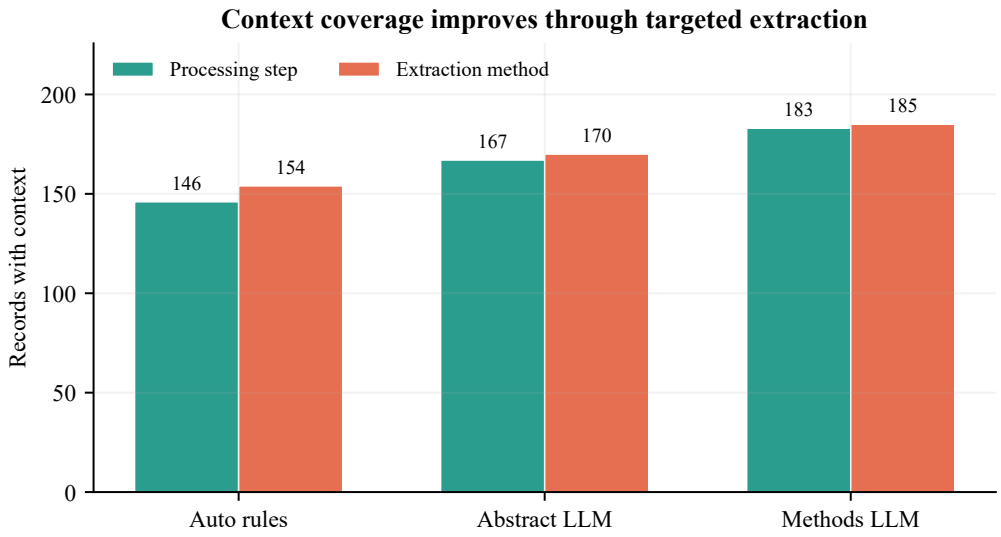

Figure 5: Context recovery across automated extraction stages. Rule-based extraction established the initial context layer, abstract-level targeted GLM5 extraction added additional context, and PMC methods-section extraction further improved recovery of processing and extraction variables.

the system creates a reproducible and updateable substrate that makes evidence level, provenance, missing context, and uncertainty visible. This framing is important because tea bioactivity claims often differ in study design, material identity, processing state, and mechanism specificity.

The results show that evidence grading changes interpretation. Tea functional activity categories contain a mixture of animal studies, human studies, meta-analyses, randomized trials, and reviews. Treating these records as equivalent would inflate confidence in claims that remain mostly preclinical. By encoding evidence level and study type as first-class graph attributes, AutoTeaKG-Silver allows users to query the evidence base under stricter assumptions.

The processing-context experiments reveal a second challenge. Processing and extraction variables are essential for interpreting tea chemistry and activity, yet many abstracts omit these details. Targeted LLM extraction and PMC methods retrieval improve coverage, but 303 records remain flagged for missing processing or extraction context. This result suggests that future improvements should focus on full-text access and methods-section extraction rather than repeatedly prompting abstracts.

The uncertainty layer is a practical compromise between scale and reliability. A manually curated gold-standard resource would be smaller and slower to update, whereas a raw LLM graph would be larger but difficult to audit. AutoTeaKG-Silver occupies the middle ground: records remain provenance-rich, uncertainty is explicit, validation results identify weak fields, and query tables and figures can be regenerated from the graph.

5 Limitations

This work has four important limitations. First, AutoTeaKG-Silver is a silver-standard resource, not a manually verified gold-standard database; users should apply uncertainty filters when drawing precise biological conclusions. Second, the current system relies on PubMed abstracts and open PMC full text, so records without accessible methods sections remain difficult to refine. Third, processing and extraction labels are normalized after LLM extraction, but some labels still represent

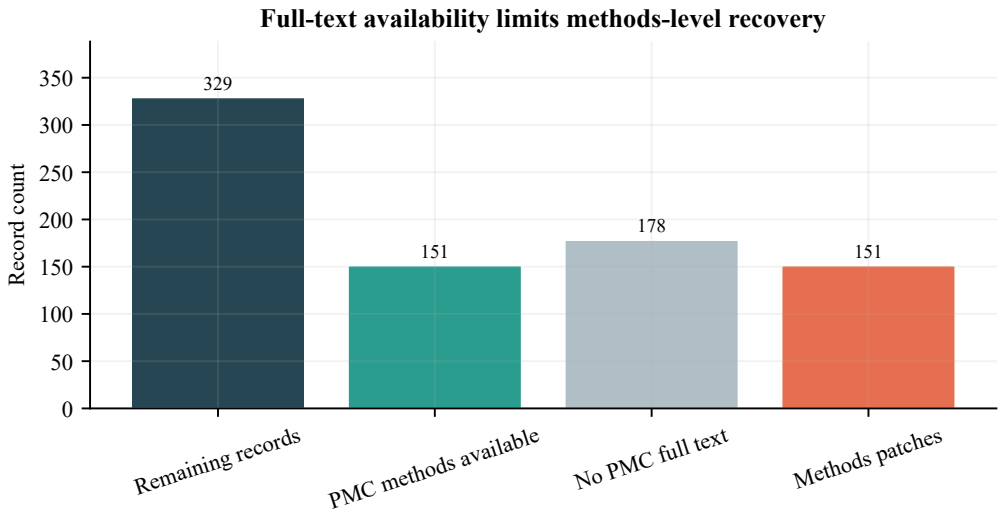

Figure 6: Full-text methods availability funnel for records that remained missing context after abstract-level extraction. Of 329 remaining records, PMC methods-like sections were available for 151 records, while 178 records lacked open PMC full text.

broad process categories rather than complete experimental protocols. Fourth, the microbiome subgraph includes grouped taxa and generic mechanism labels, so causal interpretation requires additional taxonomy normalization, full-text validation, and domain review.

6 Conclusion

AutoTeaKG-Silver demonstrates that tea functional activity literature can be converted into an automated, provenance-rich, uncertainty-aware evidence graph. The current KG v3 release contains 635 evidence records, 1,989 nodes, and 8,195 edges, and provides figure-ready query tables for evidence mapping, processing-aware analysis, uncertainty assessment, validation tracking, and microbiome mechanism exploration. The practical conclusion is deliberately bounded: automated evidence graphs can accelerate tea bioactivity synthesis when uncertainty, provenance, validation status, and missing context are treated as first-class data rather than hidden errors.

Data and Code Availability

All data products, scripts, figure-ready tables, manuscript drafts, and reproducibility artifacts used in this study are organized in the project repository. The current public repository is hosted at `git@github.com:ShawnXiha/AutoTeaKG_Silver.git`. The final KG v3 used in the manuscript is stored under `reports/methods_processing_vocab_normalized`; the main evidence table is `autoteakg_silver_records.csv`, and the graph export is provided as `kg_v3/nodes.csv` and `kg_v3/edges.csv`. Figure-ready query tables are available under `reports/kg_v3_query_tables`. SQLite database files, virtual environments, raw LLM/API response logs, and local cache files are excluded from the public repository because they are either reproducible intermediate artifacts, environment-specific files, or large raw API traces. The processed evidence tables and graph exports are included for reproducibility.

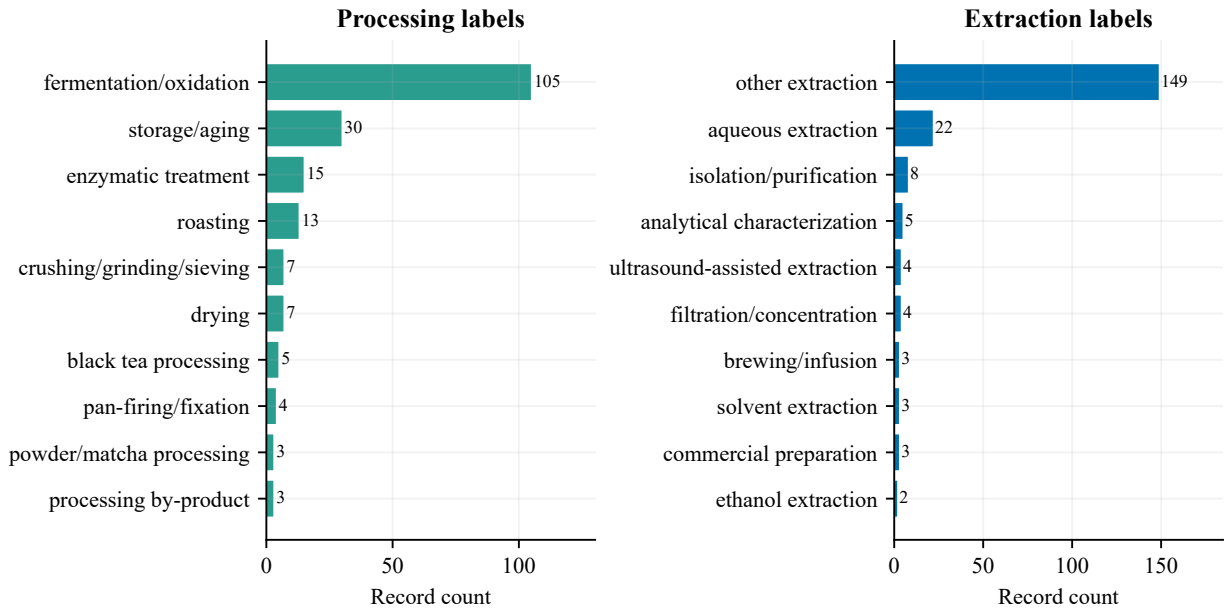

Figure 7: Top normalized processing and extraction labels after abstract-level and methods-section extraction. Vocabulary normalization reduces LLM label fragmentation by mapping surface forms such as black tea manufacturing, matcha production, brewing, isolation, and chromatographic analysis into controlled context labels.

LLM Usage Statement

This study used NVIDIA-hosted GLM5 through an OpenAI-compatible API for automatic title/abstract evidence extraction and targeted processing/extraction context recovery. LLM outputs were constrained by schema prompts, post-processed with controlled vocabularies, stored as patch files before merging, and assigned uncertainty flags. LLM extraction was not treated as manual validation. Raw LLM response logs are not included in the public repository; processed patch tables, normalized evidence records, and uncertainty metadata are included. API keys were provided only through environment variables during execution and are not stored in the repository.

Ethics Statement

This study analyzes public scientific literature, PubMed metadata, and open PMC full-text sections. It does not involve direct human-subject research, animal experimentation, clinical intervention, or access to private participant-level data. Therefore, institutional review board approval, informed consent, and animal ethics approval are not applicable.

A Supplementary Data: Reproducibility Paths

The main manuscript uses the following local release artifacts. Paths are reported relative to the project root.

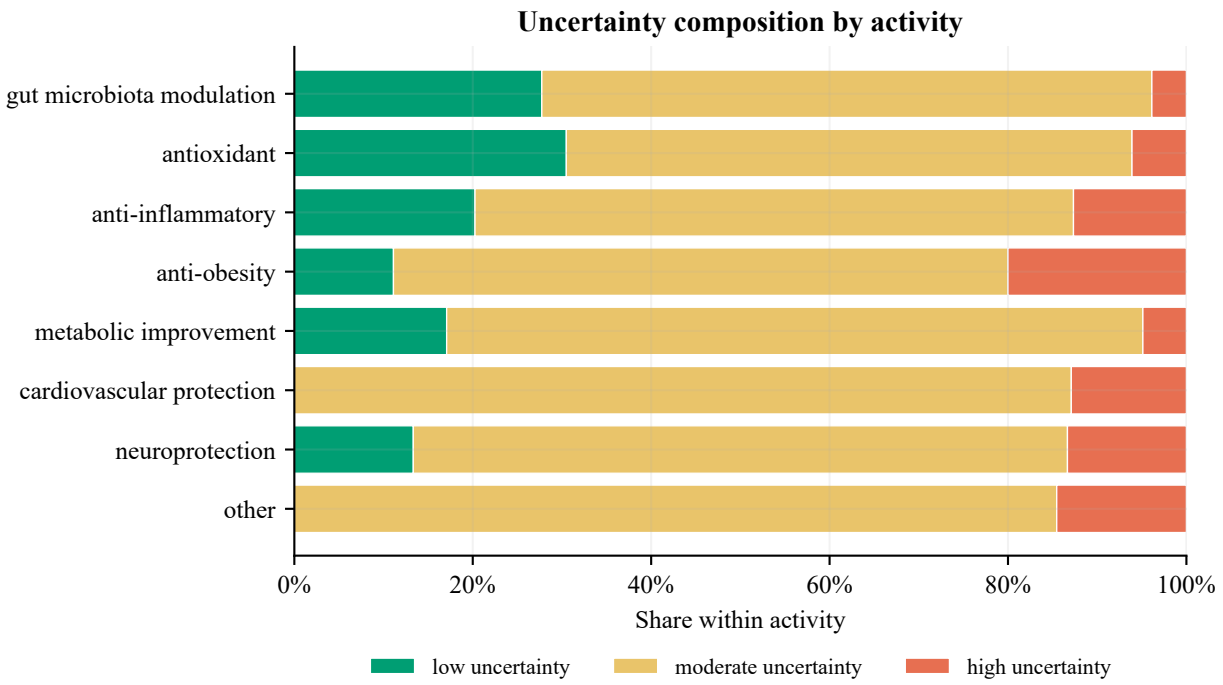

Figure 8: Uncertainty composition by activity category. Each record was assigned an uncertainty class based on LLM confidence, schema validity, mechanism specificity, context completeness, evidence level, and taxonomy status.

B Supplementary Data: Uncertainty Flag Definitions

C Supplementary Data: Validation Sample

The released worksheet evaluates activity category, study type, evidence level, component group, processing step, extraction method, and mechanism label. Field-level results should be reported only after the worksheet is completed by reviewers.

References

[1] Qiaoxian Luo, Longbiao Luo, Jinmin Zhao, Yitao Wang, and Hua Luo. Biological potential and mechanisms of tea's bioactive compounds: An updated review. *Journal of Advanced Research*, 65:345–363, 2024. doi: 10.1016/j.jare.2023.12.004.

[2] Mingchuan Yang, Xiangchun Zhang, and Chung S. Yang. Bioavailability of tea polyphenols: A key factor in understanding their mechanisms of action in vivo and health effects. *Journal of Agricultural and Food Chemistry*, 73(7):3816–3825, 2025. doi: 10.1021/acs.jafc.4c09205.

[3] Ahmed K. Rashwan, Riham A. El-Shiekh, Abdelkader Bassiony, Asem M. Abdelshafy, Ahmed I. Osman, Hala A. Younis, Fanrui Zhou, Youssuf A. Gherbawy, Hesham R. El-Seedi, Jiang Shi, and Wei Chen. Recent advances in chemical characterization, emerging applications, and biological activities of tea polysaccharides and their conjugates from leaves and

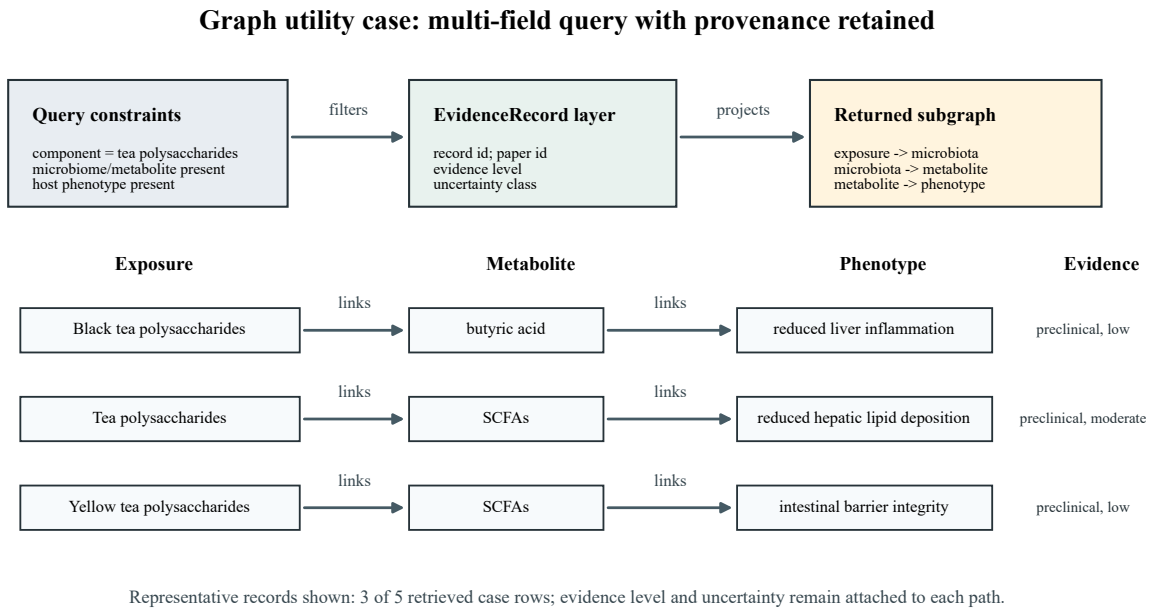

Figure 9: Graph utility case. The same query simultaneously filters component context, microbiome/metabolite evidence, host phenotype, evidence level, and uncertainty, then returns provenance-retaining evidence paths.

flowers: A review. *International Journal of Biological Macromolecules*, 328:147540, 2025. doi: 10.1016/j.ijbiomac.2025.147540.

[4] Sonali Raghunath, Sravanthi Budaraju, Seyed Mohammad Taghi Gharibzahedi, Mohamed Koubaa, Shahin Roohinejad, and Kumar Mallikarjunan. Processing technologies for the extraction of value-added bioactive compounds from tea. *Food Engineering Reviews*, 15(2): 276–308, 2023. doi: 10.1007/s12393-023-09338-2.

[5] Muhammad Aaqil, Chunxiu Peng, Ayesha Kamal, Taufiq Nawaz, Fei Zhang, and Jiashun Gong. Tea harvesting and processing techniques and its effect on phytochemical profile and final quality of black tea: A review. *Foods*, 12(24):4467, 2023. doi: 10.3390/foods12244467.

[6] Jie Wang and Zhen Li. Effects of processing technology on tea quality analyzed using high-resolution mass spectrometry-based metabolomics. *Food Chemistry*, 443:138548, 2024. doi: 10.1016/j.foodchem.2024.138548.

[7] Zimo Zhao, Ruofan Chen, and Ken Ng. Effects of differently processed tea on the gut microbiota. *Molecules*, 29(17):4020, 2024. doi: 10.3390/molecules29174020.

[8] Zheyi Song, Chi-Tang Ho, and Xin Zhang. Gut microbiota mediate the neuroprotective effect of oolong tea polyphenols in cognitive impairment induced by circadian rhythm disorder. *Journal of Agricultural and Food Chemistry*, 72(21):12184–12197, 2024. doi: 10.1021/acs.jafc.4c01922.

[9] Baoming Tian, Pinjiao Huang, Yizhu Pan, Hong Gu, Kai Yang, Zhengxun Wei, and Xiangchun Zhang. Tea polyphenols reduced obesity by modulating gut microbiota-scfas-barrier and inflammation in high-fat diet-induced mice. *Molecular Nutrition & Food Research*, 68(24): e2400685, 2024. doi: 10.1002/mnfr.202400685.

Table 6: Graph query case study for tea-polysaccharide microbiome mechanisms. The query selects records with tea-polysaccharide component context and microbiome/metabolite/host-phenotype evidence.

Record	Tea/component	Mechanism or path	Metabolite / phenotype	Evidence
41899489-R1	black tea; polysaccharides	gut-liver axis; macrophage M2 shift	butyric acid; reduced liver inflammation	preclinical; low
41899489-R2	black tea; polysaccharides	microbiota production of butyric acid	butyric acid	preclinical; low
41475763-R1	tea; polysaccharides	enrichment of SCFA producers	butyrate, acetate, propionate; reduced hepatic lipid deposition	preclinical; moderate
41475763-R2	tea; polysaccharides	microbiota-gut-liver axis signaling	butyrate, isovalerate; reduced LDL-C and hepatic lipid deposition	preclinical; moderate
41861686-R1	yellow tea; polysaccharides	barrier-related mechanism	SCFAs; intestinal barrier integrity	preclinical; low

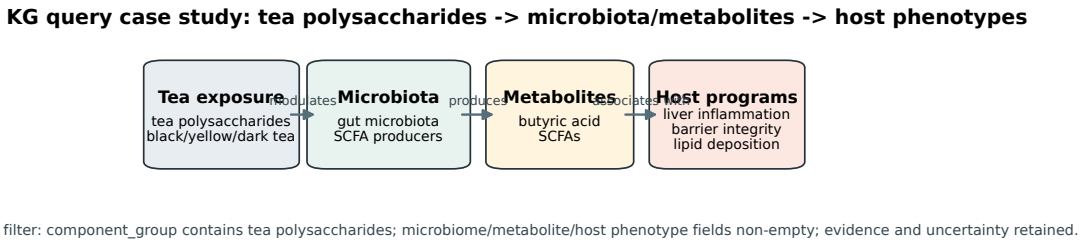

Figure 10: Graph query case study linking tea polysaccharides to microbiota/metabolite and host-phenotype nodes. The query keeps evidence level and uncertainty class attached to each evidence record.

[10] Yanlin An, Xiaoqin Zhang, Sixia Jiang, Jingjing Zhao, and Feng Zhang. Teapvs: a comprehensive genomic variation database for tea plant (camellia sinensis). *BMC Plant Biology*, 22 (1):513, 2022. doi: 10.1186/s12870-022-03901-5.

[11] Xinghai Zheng, Zahin Mohd Ali, Peng Ken Lim, Marek Mutwil, and Yuefei Wang. Comparative transcriptome database for camellia sinensis reveals genes related to the cold sensitivity and albino mechanism of 'anji baicha'. *Physiologia Plantarum*, 176(4):e14474, 2024. doi: 10.1111/ppl.14474.

[12] Yongmei Wang, Peng Wang, Yongheng Zhang, Siyi Yao, Zhipeng Xu, and Youhua Zhang. Trsrd: a database for research on risky substances in tea using natural language processing and knowledge graph-based techniques. *Database*, 2023:baad031, 2023. doi: 10.1093/database/baad031.

[13] Hui Zhao, Yuan Yang, Shuaiqi Wang, Xue Yang, Kaicheng Zhou, Caili Xu, Xuyao Zhang, Jiajun Fan, Dongyue Hou, Xingxiu Li, Ying Tan, Shanshan Wang, Xin-Yi Chu, Dongzhi Zhuoma, Fengying Zhang, Dianwen Ju, Xian Zeng, and Yu Zong Chen. Npass database

Table 7: External validation results on the 47 completed records. Acceptable = correct or minor issue among completed judgments.

Field	Completed	Acceptable	Major issues
Activity category	37	92%	2
Study type	35	80%	5
Evidence level	35	86%	3
Component group	33	94%	2
Processing step	34	100%	0
Extraction method	33	100%	0
Mechanism label	34	94%	0

Table 8: Key reproducibility artifacts for the current manuscript release.

Role	Relative path
Final KG v3 directory	`reports/methods_processing_vocab_normalized`
Final silver record table	`reports/methods_processing_vocab_normalized/autoteakg_silver_records.csv`
Final KG node table	`reports/methods_processing_vocab_normalized/kg_v3/nodes.csv`
Final KG edge table	`reports/methods_processing_vocab_normalized/kg_v3/edges.csv`
Figure-ready query tables	`reports/kg_v3_query_tables`
PubMed search archive	`reports/pubmed_search_archive_2026-04-02.md`
Silver KG construction	`scripts/build_autoteakg_silver_v1.py`
Targeted processing extraction	`scripts/targeted_processing_llm_extractor.py`
PMC methods retrieval	`scripts/retrieve_pmc_methods_for_remaining_context.py`
Methods extraction batches	`scripts/run_methods_processing_extraction_batches.py`
Vocabulary normalization	`scripts/normalize_processing_extraction_vocab.py`
Figure and table generation	`scripts/generate_kg_v3_query_tables_and_figures.py`

update 2023: quantitative natural product activity and species source database for biomedical research. *Nucleic Acids Research*, 51(D1):D621–D628, 2023. doi: 10.1093/nar/gkac1069.

[14] Sanya B. Taneja, Tiffany J. Callahan, Mary F. Paine, Sandra L. Kane-Gill, Halil Kilicoglu, Marcin P. Joachimiak, and Richard D. Boyce. Developing a knowledge graph for pharmacokinetic natural product-drug interactions. *Journal of Biomedical Informatics*, 140:104341, 2023. doi: 10.1016/j.jbi.2023.104341.

[15] Ruyi Zhang, Qiling Li, Yuxuan Gu, and Wenli Liao. Harnessing the power of fermented tea to improve gut microbiota and combat obesity epidemic. *Biology*, 13(10):779, 2024. doi: 10.3390/biology13100779.

[16] Beatriz Arce-López, Guadalupe X. Bazan, Susana Molina, Maria Carmen Crespo, María García-Beccaria, Silvia Cruz-Gil, Cristina M. Fernández-Díaz, Ana Ramírez de Molina, Ricardo Ramos-Ruiz, and Maria Isabel Espinosa-Salinas. Effect of fiber-modified kombucha tea on gut microbiota in healthy population: A randomized controlled trial (rct). *Current Research in Food Science*, 11:101130, 2025. doi: 10.1016/j.crfs.2025.101130.

Table 9: Uncertainty flags used in AutoTeaKG-Silver. Flags are stored on evidence records and propagated to graph edges through the evidence-record node.

Flag	Trigger	Interpretation	Severity
`low_llm_confidence`	Original LLM confidence score is below 0.80.	The extracted evidence record should be prioritized for review.	severe
`taxonomy_expansion_needed`	The normalized activity category is `other`.	The record is either out of scope or requires activity-taxonomy expansion.	severe
`missing_component_context`	No component group is available after rule and LLM context extraction.	The record lacks a key exposure descriptor.	severe
`missing_processing_or_extraction_context`	Both processing step and extraction method are empty.	The record cannot support processing-aware comparison.	moderate
`generic_mechanism`	Mechanism label is empty or belongs to a generic placeholder list.	The record supports an activity claim but weakly supports mechanistic graph reasoning.	severe
`missing_named_microbiota`	Activity is gut microbiota modulation but no named taxon is extracted.	Microbiome interpretation should remain generic.	moderate
`preclinical_only`	Study type is animal study.	The claim should not be interpreted as direct human evidence.	moderate
`review_summary_record`	Study type is systematic review.	The record summarizes literature rather than representing one primary experiment.	moderate
`low_evidence_level`	Evidence level is low preclinical or in vitro.	The claim has weak translational support.	moderate
`uncertain_effect_direction`	Effect direction is unclear, no clear effect, or mixed.	The extracted claim should not be counted as straightforward positive support.	moderate
`missing_abstract`	No abstract text is available.	The record has reduced text evidence for automatic extraction.	moderate
`invalid_*`	A controlled-vocabulary field contains an invalid label.	The record violates schema constraints and requires normalization or exclusion.	severe

Table 10: External validation sample design. The sample is stratified by uncertainty class and is released as a worksheet for field-level review. Validation is not used to construct KG v3.

Uncertainty class	Sampled records	Fields reviewed
Low uncertainty	14	7
Moderate uncertainty	24	7
High uncertainty	10	7
Total	48	7